

Herpesvirus Infections

Cheryl Newman-Whitlow, MD • 66-slide current deck • HSV → VZV → EBV → CMV → HHV-6/7/8 → Mpox

4 official objectives

3 ScholarRx bricks

CSF reasoning

VZV vaccines

Mpox overview

SOURCE PRIORITY: the current 66-slide Newman-Whitlow deck (Sept 8, 2026) controls terminology, examples, treatment/prevention statements, and faculty emphasis. The matching MCG 2029 AI Notes handout, “Lecture 7: Herpesvirus Infections”, is secondary enrichment. Board-style mechanisms, latency details, and classic buzzwords from the handout are labeled when they are not stated on the current deck. When the handout conflicts with the 2026 deck, the deck wins.

Official Week 8 Objectives

1. Describe primary clinical features, CSF findings, and complications of herpesvirus infections in normal and immunocompromised hosts.

2. Explain treatment and prevention strategies, including medications and VZV vaccines.

3. Distinguish herpesviral disease from other pathogens using CSF analysis.

4. Discuss U.S. epidemiology, clinical manifestations, diagnostic tests, and prevention of Mpox.

Herpes Simplex Infections & Genital Warts

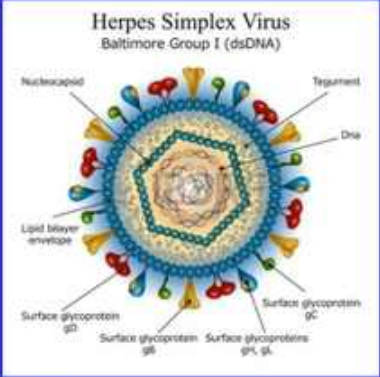
Varicella-Zoster Virus

Drugs to Treat Herpesvirus Infections

Slides 2-9: Herpesvirus Family Logic

Herpesvirus Biology

- Large **dsDNA** viruses
- Four structural features: outer **envelope**, tegument, nucleocapsid, inner DNA core
- Important surface and tegument glycoproteins mediate cell entry and viral replication.



Slide 5. Herpesviruses are large dsDNA viruses with envelope, tegument, nucleocapsid, and inner DNA core. Surface/tegument glycoproteins mediate entry and replication.

FAMILY IDENTITY

Herpes = enveloped dsDNA + latency + reactivation. The envelope makes herpesvirus fragile outside the body.

Viral biology

- **Subfamilies -**
 - Alpha** - HSV1&2, VZV
 - Beta** – CMV, HHV6 and 7
 - Gamma** – EBV, HHV8
- **Latency and reactivation**
 - **Latency** - no infectious virions are produced
In HSV 1/2 *neuronal* cell is primary latent site; for others may be lymphoid cells
 - **Reactivation** – many viral/host factors play a role including tegument proteins

Slide 8. Alpha = HSV-1/2, VZV; Beta = CMV, HHV-6/7; Gamma = EBV, HHV-8. The current deck emphasizes latency and reactivation across the family.

Subfamily	Members	Clinical memory
Alpha	HSV-1, HSV-2, VZV	Neurotropic latency / recurrent mucocutaneous disease
Beta	CMV, HHV-6, HHV-7	Congenital/opportunistic CMV; roseola syndromes
Gamma	EBV, HHV-8	Lymphoid latency / oncogenic associations

Handout-only enrichment: the 2029 handout adds that herpesviruses have **linear dsDNA, an icosahedral capsid, and nuclear replication**. It also gives the classic latency sites: HSV-1 trigeminal ganglia, HSV-2 sacral ganglia, VZV dorsal-root ganglia, EBV B cells, and CMV mononuclear cells.

Slides 10-11: General Laboratory Diagnosis

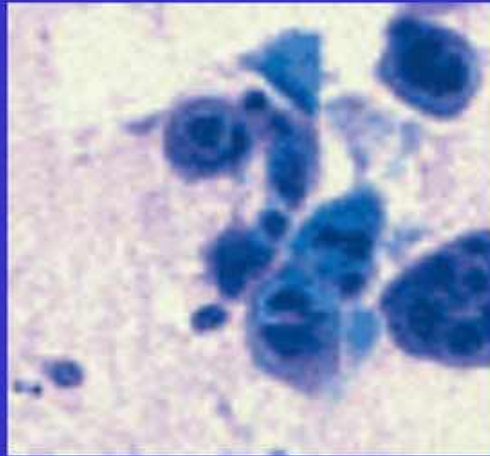
LABORATORY DIAGNOSIS

- **Tissue (cell) culture** isolation-HSV,VZV, CMV
- **Antigen detection** - ELISA or IF (immunofluorescence): CMV,HSV,VZV
- **Nucleic acid detection** (PCR): identification of virus in blood, CSF and other body fluids
- **Serology** – IgM (acute infection); IgG prior exposure based on Ab to gpG
- **Histopathology** -Immunohistochemistry
Cytopathology: **Tzanck** smear – Multinucleated giant cells



Slide 10. The deck lists culture, antigen detection, PCR from blood/CSF/other fluids, serology, histopathology/immunohistochemistry, and Tzanck cytopathology.

Tzanck Prep

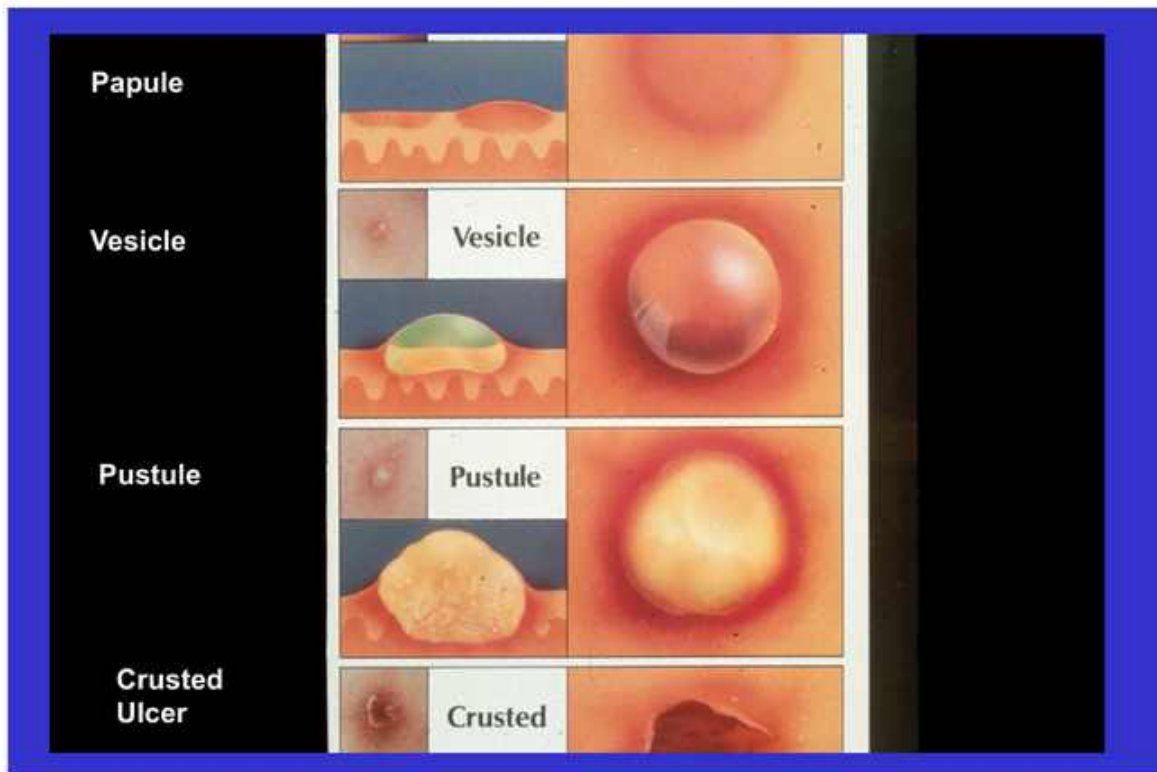


Slide 11. Tzanck preparation demonstrates multinucleated giant cells. Useful morphologic clue, but not virus-specific.

Method	Deck use
PCR / nucleic acid detection	Identify herpesviral DNA in blood, CSF, or other body fluids; especially important for HSV encephalitis
Serology	IgM suggests acute infection; IgG prior exposure
Tzanck smear	Multinucleated giant cells
Histopathology	CMV owl's-eye and tissue-specific disease confirmation

Tzanck trap: multinucleated giant cells support HSV/VZV, but they do not distinguish HSV-1 from HSV-2 from VZV. PCR gives etiologic identification.

Slides 12-23: Herpes Simplex Virus 1 & 2



Slide 13. Lesion evolution shown visually: papule → vesicle → pustule → crusted ulcer.

HSV-1	HSV-2
Primary infection often childhood; saliva transmission; gingivostomatitis in children or exudative pharyngitis in adults	Painful grouped genital vesicles on erythematous base, progressing to pustules/ulcers
Latency usually trigeminal ganglion in current deck	Recurrent genital disease; prolonged asymptomatic shedding may occur
Herpetic whitlow, herpes gladiatorum, keratitis, encephalitis, eczema herpeticum	First episode may include fever, adenopathy, myalgia, headache, aseptic meningitis

Primary: HSV (subtype) 1 Infection

- Usually acquired in **childhood**; Tx: **saliva**
- **Most primary HSV1 is asymptomatic**
- Incubation 2 - 12 days
 - **Gingivostomatitis** in children
 - painful blisters/ulcers involving lips/gums/mouth – dehydration
 - **Exudative pharyngitis** – adults
- Resolves in 10 to 14 days; **establishes latency** in sensory ganglia (usually trigeminal)



Slide 14. Primary HSV-1: painful oral lesions / gingivostomatitis, then latency in sensory ganglia, usually trigeminal.

Recurrent HSV 1 infections

- **Due to reactivation** – clinical episodes of productive viral replication with spread from neural cell to axon and then skin
- **Prodrome is frequent**
- **Provoked by:**
fever, stress, sun, trauma,
decreased immunity,
steroid therapy
- **Fever blisters/cold sores**



Slide 16. Recurrent HSV-1 travels from latent neural site back to skin. Triggers include fever, stress, sun, trauma, decreased immunity, and steroids.

HSV 2 Clinical Manifestations

- Incubation: 2-12 days (but can be longer)
- ***Painful Grouped vesicles on erythematous base***
- Progression to **pustules** then **ulcer**
- Location: external **genitalia**; cervix, urethra (urinary retention), anus; oral/skin; rare meninges
- With **first episode systemic symptoms** may occur- fever, adenopathy, myalgia, HA or aseptic meningitis; more severe in women
- Viral shedding- usually 11days/ heals 2-4 weeks; ***prolonged asymptomatic shedding may occur***

Slide 17. HSV-2 clinical manifestations: painful grouped vesicles, genital/urethral/anal disease, systemic symptoms in first episode, and asymptomatic shedding.

Treatment of Herpes Infections

- **Initial Treatment:**
Acyclovir or
Famciclovir or
Valacyclovir
- **Recurrences:** above/shorter courses
- **Suppression – usually in immunocompromised:**
Acyclovir or
Famciclovir or
Valacyclovir

Slide 20. Current deck treatment: acyclovir, famciclovir, or valacyclovir for initial episodes, recurrences, and suppression.

HSV encephalitis: must-not-miss vignette

HSV Complications *Herpes Encephalitis*

- Due to **HSV 1 reactivation**
–trigeminal ganglion
- Neural extension to the **temporal lobe**
- **HA, fever, behavior changes, seizures**
- CSF: mixed **pleocytosis**, **↑protein**,
↑RBCS
- DX: **HSV PCR of CSF** or brain biopsy
- RX: **Acyclovir IV 10 mg/kg q 8 hours**
- Mortality: **20% despite treatment**



necrotizing hemorrhagic encephalitis

Slide 21. HSV-1 encephalitis: temporal-lobe predilection, fever/headache/behavior change/seizures, mixed CSF pleocytosis with elevated protein and RBCs, diagnosis by CSF HSV PCR, IV acyclovir.

HSV ENCEPHALITIS

Temporal lobe + behavior/seizures + RBCs in CSF → HSV-1 until proven otherwise → CSF HSV PCR + IV acyclovir.

HSV Complications

- **Neonatal herpes** – meningoencephalitis, disseminated or localized disease in baby
- **Ocular herpes**
– keratitis or retinitis
- **Herpes reactivation in immunocompromised** – esophagitis, organ involvement



Slide 22. Additional complications include neonatal herpes, ocular keratitis/retinitis, and reactivation with esophagitis or organ disease in immunocompromised hosts.

Eczema herpeticum – HSV occurring in setting of active skin disease (atopic dermatitis, eczema, etc)



Slide 23. Eczema herpeticum: HSV occurring in the setting of active atopic dermatitis/eczema.

Handout-only enrichment: HSV-1 classically travels by retrograde axonal transport to the trigeminal ganglion, HSV-2 to sacral ganglia; HSV keratitis produces branching dendritic ulcers. The handout also adds erythema multiforme as a possible HSV-triggered hypersensitivity reaction.

Slides 24-33: Varicella-Zoster Virus

ONE VIRUS, TWO DISEASES

Primary VZV = varicella/chickenpox. Reactivation = zoster/shingles.

Varicella Zoster Virus

- **Primary varicella (first exposure)**
 - **Chicken pox** in children/more serious in adults
 - **Disseminated varicella**-immunocompromised
- **Recurrences (reactivation – prior exposure):**
 1. **Shingles** (herpes zoster)- usually dermatomal rash
Post-herpetic neuralgia
 2. **Disseminated zoster**- immunocompromised

Slide 24. Primary varicella is chickenpox; reactivation produces dermatomal zoster. Immunocompromised patients can develop disseminated disease.

Acute Varicella – *Chicken Pox*

- Respiratory transmission or contact w/active lesions
- 10-14 day incubation
- 1 or 2 day prodrome of fever, HA, malaise, anorexia
- **Rash usually begins on trunk/occurs in crops over several days due to cyclic viremia**
- **Lesions progress through stages: vesicle → pustule → crusted ulcer**
- **Pruritic**
- **Contagious until all lesions crust**

Slide 25. Chickenpox: respiratory/contact transmission, 10-14 day incubation, rash in crops from cyclic viremia, lesions in multiple stages, contagious until all crusted.

Chickenpox – healthy child



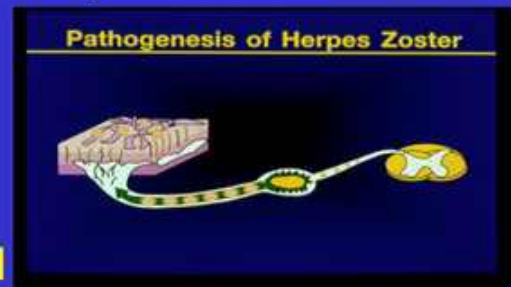
Slide 26. Healthy child with chickenpox - the deck visually reinforces lesions in different stages.



Slide 27. Disseminated varicella in an immunocompromised adult.

Herpes zoster - Shingles

- Reactivation of latent varicella virus from a sensory ganglion
- Usually occurs years after initial exposure
- Painful dysesthesia followed by **dermatomal** vesicular rash
- **Pain: symptom/burning**
- **Thoracic and lumbar dermatomes-
most frequently affected**



Slide 29. Shingles: reactivation from sensory ganglion, painful dysesthesia followed by dermatomal vesicular rash; thoracic/lumbar dermatomes are common.

Post Herpetic Neuralgia(PHN)

- **Persisting pain** in dermatome **after** resolution of shingles rash
Most common late complication-occurs in 10-15%
- Common in **elderly**; over 50% are > age 50
- Probably due to structural damage to nerves
- **Early treatment** of shingles may reduce **occurrence** and **duration** of PHN

Slide 31. Post-herpetic neuralgia = persistent dermatomal pain after rash resolution; the deck calls it the most common late complication.

Situation	Current-deck management / prevention
Varicella high-risk exposure	VZIG
Varicella vaccine	VARIVAX for childhood prevention and susceptible adults
Zoster	Famciclovir or valacyclovir; IV acyclovir if immunocompromised; start within 3 days
Shingles prevention	Shingrix recombinant subunit vaccine, 2 doses; healthy adults >50 and immunocompromised adults >18 per lecture

Prevention of Shingles

Shingrix Vaccine

- Recombinant subunit vaccine (with adjuvant)
- Provides vigorous immune response – Elderly (70-80yo) have >90% protection~4 yrs
- **Two** dose series- 2nd shot 2-6 months after 1st
- Recommended: Healthy adults >50 yrs and immunocompromised >18 years
- Can give even if hx of prior shingles, Zostavax (**old live viral vac**) or unclear varicella hx

Slide 33. Shingrix: recombinant subunit vaccine with adjuvant; two-dose series; lecture recommendation includes healthy adults >50 and immunocompromised adults >18.

Handout-only enrichment: the 2029 handout emphasizes dorsal-root ganglion latency, shingles typically not crossing the midline, and herpes zoster ophthalmicus with Hutchinson sign as an emergency. Those details are useful board enrichment but are not all written on the current deck.

Slides 34-41: Epstein-Barr Virus

Epstein Barr Virus(EBV)

- Infectious Mononucleosis – primary EBV
- Burkitt's lymphoma
- **Other** lymphomas in immunodeficiency (esp AIDS)
- Nasopharyngeal carcinoma
- EBV lymphoproliferative syndromes- post transplant
- Oral hairy leukoplakia in AIDS

Slide 34. EBV disease spectrum in the current deck: infectious mononucleosis, Burkitt lymphoma, other immunodeficiency-associated lymphomas, nasopharyngeal carcinoma, post-transplant lymphoproliferative syndromes, and oral hairy leukoplakia.

Clinical Features of Mono (IM)

- Asymptomatic **to** life-threatening
- **SYMPTOMS**: malaise and fatigue; sore throat
- **SIGNS**: fever, pharyngitis (with palatal petechiae), **adenopathy**, splenomegaly (50%), hepatomegaly(12%)
- Rash after receipt of aminopenicillin

Slide 37. Mono: fatigue/malaise, sore throat, fever, pharyngitis with palatal petechiae, adenopathy, splenomegaly, hepatomegaly, and aminopenicillin-associated rash.

Clinical Course of IM

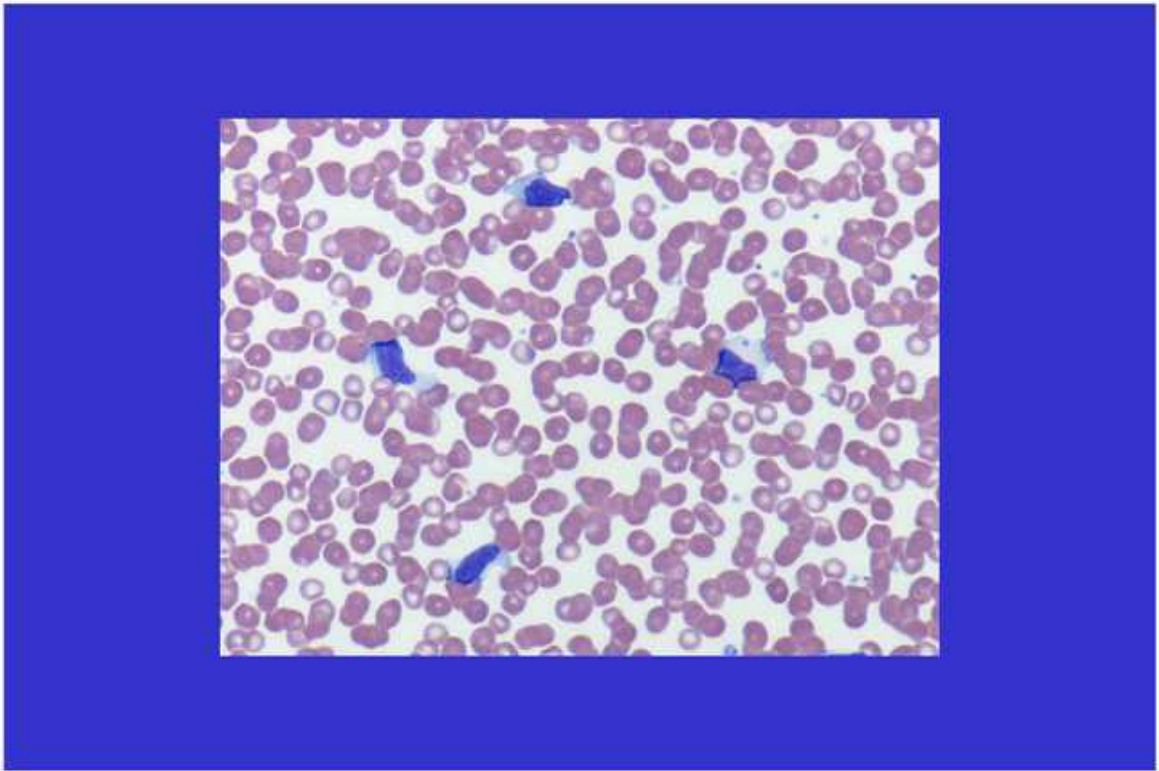
- **Sore throat** 3-5 days
- **Fever** 10-14 days
- **Prostration** 2-3 weeks, may wax and wane
- **LABORATORY DIAGNOSIS:** Mild leukocytosis with relative neutropenia

Atypical lymphocytosis(>10%) – T cells (CD8)

Positive heterophile Ab – MONOSpot test

IgM VCA and EA seen in acute disease

Slide 39. Laboratory diagnosis: mild leukocytosis, atypical lymphocytosis (>10%) consisting of CD8 T cells, positive heterophile antibody/Monospot, and acute VCA/EA IgM patterns.



Slide 40. Peripheral smear image of atypical/reactive lymphocytes in infectious mononucleosis.

EBV MONO

Fever + pharyngitis + adenopathy + splenomegaly + heterophile positive + atypical CD8 T cells.

Management: rest and no contact sports. The deck states that no specific antiviral therapy is beneficial; steroids may be used for selected serious complications such as hemolytic anemia, severe thrombocytopenia, impending airway obstruction, or CNS/ cardiac disease.

Handout-only enrichment: EBV binds CD21 on B cells; the atypical “Downey cells” are reactive CD8 T cells, not infected B cells. The handout also emphasizes posterior cervical lymphadenopathy and the heterophile antibody's agglutination principle.

Slides 42-49: Cytomegalovirus

CMV Infection

- **Transmitted by close contact** with infected secretions – saliva, urine, semen, blood, milk
- Many exposed in childhood – daycares
- **Primary Syndromes:**
 - ***CMV Mononucleosis**
 - ***Congenital CMV infection**
 - ***CMV Opportunistic Disease-immunosuppressed**

Slide 42. CMV transmission occurs through close contact with infected secretions; major primary syndromes are CMV mononucleosis, congenital CMV, and opportunistic disease in immunosuppressed hosts.

CMV Mononucleosis Syndrome Primary CMV

- Healthy young adults **Fever, malaise, adenopathy, enlarged spleen;**
- **Atypical lymphocytosis**
- **Negative heterophile Ab**
- Rare complications: hemolytic anemia, thrombocytopenia
- Course may be prolonged
- No specific Rx in healthy host
- **Clinically Mimics EBV Mononucleosis**

Slide 43. CMV mononucleosis in healthy young adults can mimic EBV mono but is heterophile/Monospot negative.

Congenital CMV Disease

- **Cytomegalic Inclusion Disease** – congenital infection caused by primary CMV in mother
TORCH syndrome*
Jaundice, hepatosplenomegaly, rash, microcephaly, cerebral calcification, chorioretinitis, seizures
- **Perinatal infection**: due to acquisition at birth or from breast feeding
Mild or no symptoms – but may shed virus in secretions (urine) for months

*TORCH: *Toxoplasma*, (O-other) *Rubella*, *CMV*, *HSV*

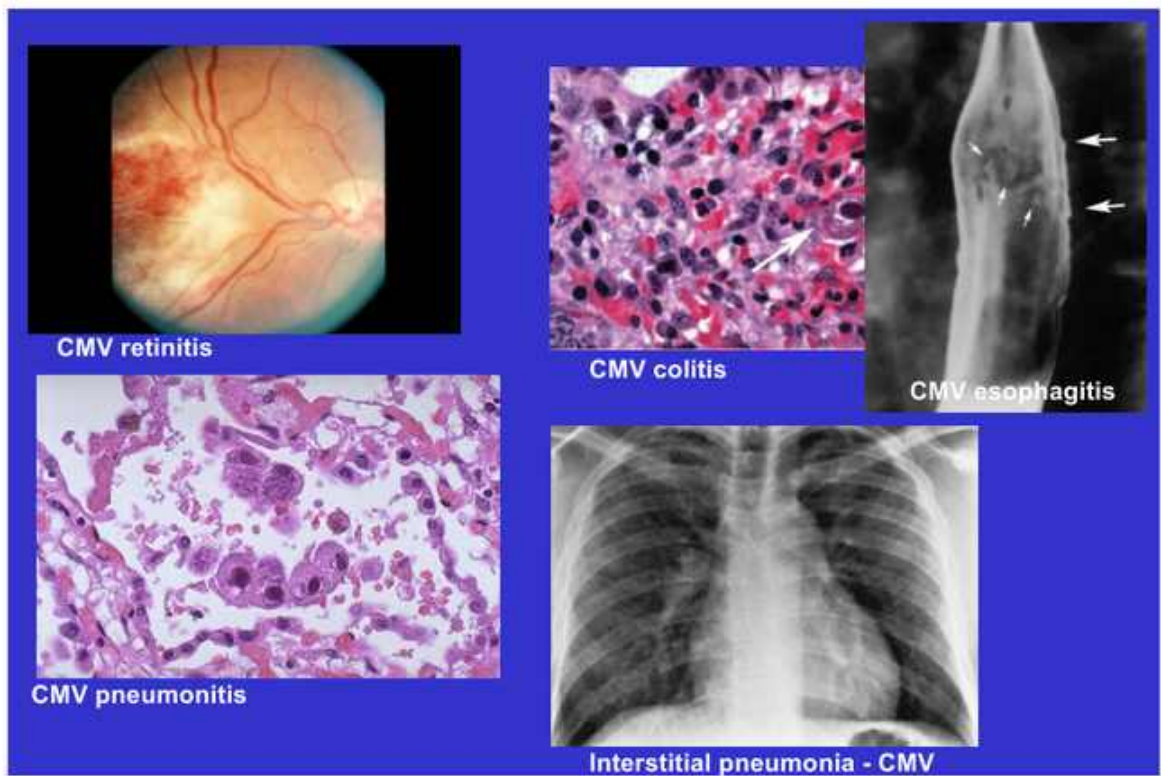
Slide 44. Congenital CMV / TORCH pattern in the current deck includes jaundice, hepatosplenomegaly, rash, microcephaly, cerebral calcification, chorioretinitis, and seizures.

CMV Opportunistic Disease

- **Immunosuppressed**: HIV, **BMT***, **Organ Transplant** or chemotherapy
- **Reactivation or Primary Infection**
- **Highest risk: 1-4 months post TX** (transplant)
- **Clinical: *Fever and Leukopenia***
Direct organ involvement:
retinitis, pneumonitis, encephalitis, esophagitis, colitis, myelitis, polyradiculitis, rare skin or mucosal lesions
- **May be associated with graft (transplanted organ rejection) loss; high mortality**

*Bone Marrow Transplant

Slide 45. Opportunistic CMV: HIV, transplant, BMT, chemotherapy; organ disease includes retinitis, pneumonitis, encephalitis, esophagitis, colitis, and neurologic disease.



Slide 46. The deck visually shows CMV retinitis, colitis, esophagitis, pneumonitis, and interstitial pneumonia.



Owl's eye inclusion

Slide 48. Characteristic CMV owl's-eye inclusion on histopathology.

Treatment of CMV Infection

- **Ganciclovir** – IV
- **Valganciclovir** – PO
- Foscarnet
- Cidofovir
- CMV immune globulin
- **Letermovir** – prevention of reactivation in BMT
- Maribavir – for resistant CMV in Tx patients
- **NO Available Vaccine**

Slide 49. Current-deck treatment list: ganciclovir, valganciclovir, foscarnet, cidofovir, CMV immune globulin, letermovir for prevention in BMT, maribavir for resistant CMV; no vaccine.

EBV vs CMV mononucleosis

Feature	EBV	CMV
Heterophile / Monospot	Positive	Negative
Atypical lymphocytes	Yes	Yes
Pharyngitis / marked adenopathy	Prominent in deck	Can occur, but deck emphasizes mono-like mimic
Special host	Teen/young adult classic IM	Congenital disease; severe opportunistic disease in transplant/HIV

Ozzy checkpoint - click him once you can separate EBV mono from CMV mono and then pivot to the correct confirmatory test.

Handout-only enrichment: congenital CMV board clues include sensorineural hearing loss and periventricular calcifications; CMV retinitis may have a “pizza-pie / ketchup-and-mustard” appearance. The handout also adds ganciclovir marrow suppression and foscarnet/cidofovir nephrotoxicity - mechanisms/toxicities are not detailed on the current deck.

Slides 50-54: HHV-6, HHV-7, HHV-8

HHV 6 – Exanthem subitum

- **Roseola** infantum – HHV6B or **Sixth** disease
- Incubation period: 5-10 days
- Acute onset **high** fever
- **Defervescence and maculopapular rash**
- Resolves in 1-2 days
- Rarely causes neurologic symptoms – febrile seizures or encephalitis

Slide 51. HHV-6B roseola: abrupt high fever followed by defervescence and maculopapular rash; febrile seizures/encephalitis are rare neurologic complications.



Slide 52. Roseola image - rash after the fever breaks is the key sequence.

ROSEOLA

High fever first → fever resolves → rash appears.

HHV-7: acquired in childhood, saliva transmission, syndromes similar to HHV-6; can reactivate in BMT/SOT.

HHV 8: Kaposi sarcoma (associated) herpesvirus- KSHV

- **Cause of Kaposi sarcoma** – vascular multicentric tumor found in HIV patients
- Transmission: sexual and probably other (saliva)



- **Primary Effusion Lymphoma (PEL)**: rare lymphoma seen in HIV involving body cavities
- **Multicentric Castleman's Disease**: rare lymphoproliferative disease- fever, enlarged liver/spleen and adenopathy

Slide 54. HHV-8/KSHV: Kaposi sarcoma, primary effusion lymphoma, and multicentric Castleman disease are the current-deck associations.

Handout-only enrichment: HHV-8 drives endothelial proliferation and angiogenesis in Kaposi sarcoma; the handout describes violaceous mucocutaneous lesions and AIDS association.

Slides 55-62: Faculty Questions = Exam Logic

Question 1

- A 65 yo male presents with 1 day history of fever, HA, confusion and behavioral changes. CSF shows: WBC: 1800; RBCs 1200, Protein 200; India ink and gram stain are negative and MRI shows inflammation in the temporal lobe. The diagnosis is best confirmed by:
 - A. Serum cryptococcal Ag
 - B. CSF HSV1 PCR
 - C. Viral culture of the spinal fluid
 - D. Herpesvirus IgM
 - E. CMV PCR of blood

Slide 55. Question 1: acute fever/headache/confusion + RBC-rich CSF + temporal-lobe inflammation. The intended confirmation is CSF HSV-1 PCR.

CSF / clinical pattern	Interpretation from this lecture
Temporal lobe + behavior change/seizures + RBCs + ↑protein	HSV-1 encephalitis → CSF HSV PCR
Teen + sore throat + adenopathy + positive heterophile test	EBV IM → excess reactive T lymphocytes
Painful genital vesicles improving with acyclovir	HSV recurrence remains possible , often less severe
Which herpesvirus illness is vaccine-preventable?	Acute varicella

Question 2

- A 15 yo male presents with sore throat, swollen lymph nodes, fatigue and fever. The heterophile antibody is positive. His “CBC with diff” will likely reveal:
 - A. An excess of PMN's
 - B. An excess of B lymphocytes
 - C. An excess of platelets
 - D. An excess of basophils
 - E. An excess of T lymphocytes

Slide 57. Question 2 reinforces that atypical lymphocytes in EBV mononucleosis are T lymphocytes, not B lymphocytes.

Question 3

- An 18 yo female presents with a painful vesicular eruption in the genital area. You prescribe Acyclovir and she improves. You counsel her that:
 - A. The disease may recur but will likely be less severe
 - B. She is likely “cured” but make sure her partners wear condoms
 - C. She should abstain from sex indefinitely
 - D. She might be at risk for post-herpetic neuralgia
 - E. She should not get pregnant because of risk gestational herpes

Slide 59. Question 3 reinforces latency/recurrence counseling after genital herpes.

Question 4

- Which of the following herpesvirus related illnesses is preventable by vaccination?
- A. Infectious mononucleosis
 - B. Herpes simplex encephalitis
 - C. Monkeypox
 - D. Acute varicella
 - E. Roseola

Slide 61. Question 4 reinforces that acute varicella is vaccine-preventable.

Slides 63-66: Brief Mpox Overview

Monkeypox -- Mpox

- **Orthopoxvirus** – (smallpox, cowpox); zoonotic
- Identified in infant 1958 in DRC – circulates in central and west Africa
- **Transmission: direct skin to skin** contact with infected lesions; **In US – primarily genital**
- Incubation: about 1 week (6-17 days)
- Clinical: **flu like prodrome** and rash often presenting simultaneously
-painful vesicles often at inoculation site – ie genital; progresses thru stages to crust

Slide 63. Mpox is an orthopoxvirus, not a herpesvirus. Current deck emphasizes direct skin-to-skin transmission from lesions, often genital in U.S. cases, ~1 week incubation, flu-like prodrome, and painful vesicles progressing to crust.



Slide 64. CDC visual examples of Mpox lesions included in the deck.

Mpox – Dx, Rx and Px

- Dx: **PCR** of lesional fluid
- Rx: No known therapy; **Tecovirimat (TPOXX)** – antiviral for smallpox failed in trials of MPOX
- Px: **JYNNEOS**- live modified vaccinia (orthopoxvirus) strain - non replicating attenuated
Prevents Smallpox and Mpox
2 dose series – IM/ID
Currently exposed or at risk (esp sexual exposure) are eligible
- Mortality: ~ 4-11% (c/w smallpox – 30%)

Slide 66. Current 2026 deck: diagnosis by lesion PCR; JYNNEOS is a non-replicating modified vaccinia vaccine, 2-dose IM/ID series, used for people exposed or at risk.

Important source conflict: the older MCG 2029 handout says tecovirimat can be used for severe Mpox. The **current 2026 Newman-Whitlow deck explicitly says there is no known therapy and that tecovirimat failed in Mpox trials.** This companion follows the current deck.

High-Yield Traps & Vignette Recognition

If you see...	Think...
Temporal-lobe encephalitis + RBCs in CSF	HSV-1 → CSF PCR + IV acyclovir
Painful grouped genital vesicles + recurrence	HSV-2 / genital HSV
Multiple lesion stages in crops	Varicella
Unilateral dermatomal painful vesicles	Zoster
Persistent pain after zoster rash resolves	Post-herpetic neuralgia
Teen mono + heterophile positive + atypical T cells	EBV
Mono-like illness + heterophile negative	CMV
Transplant/HIV + retinitis/colitis/esophagitis + owl's eye	CMV opportunistic disease
High fever that resolves before rash	HHV-6/7 roseola
Kaposi / PEL / multicentric Castleman	HHV-8
Painful inoculation-site lesions + skin-to-skin exposure	Mpox → lesion PCR

LAST 60 SECONDS

HSV = neurotropic recurrence. VZV = chickenpox then shingles. EBV = heterophile-positive mono. CMV = heterophile-negative mono + congenital/opportunistic disease. HHV-6/7 = fever then rash. HHV-8 = Kaposi. Mpox is orthopox, not herpes.

Source hierarchy: current 66-slide Newman-Whitlow deck is primary. Official Week 8 objectives determine scope and ScholarRx. MCG 2029 AI Notes "Lecture 7: Herpesvirus Infections" provides secondary board-style enrichment only.